

Principles of Congestion Control

(Causes, Costs, and Network Scenarios)

1. What is Network Congestion?

Congestion occurs when too many sources attempt to send data at too high a rate for the network to handle. While reliable data transfer treats the *symptom* (packet loss), congestion control treats the *cause* by throttling senders.

The "Top-Ten" Importance

Congestion control is one of the most fundamental problems in networking, impacting everything from interactive web performance to large-scale data center throughput.

2. The Causes and Costs of Congestion

We examine three scenarios to understand why congestion occurs and the resources it wastes.

2.1 Scenario 1: Two Senders, Infinite Buffers

- **Setup:** Two hosts (A and B) share a single hop with link capacity R .
- **Assumption:** The router has **infinite buffers**; no packets are ever dropped.

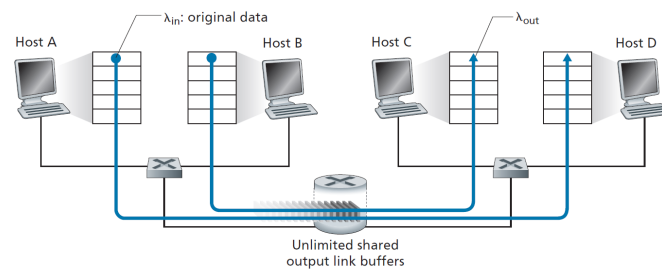


Figure 1: Scenario 1: Infinite Buffers

Performance Analysis

- **Throughput:** For sending rate $\lambda_{in} \leq R/2$, throughput $\lambda_{out} = \lambda_{in}$. For $\lambda_{in} > R/2$, throughput stays at $R/2$.
- **Delay:** As arrival rate approaches capacity, queuing delay increases dramatically. At $\lambda_{in} > R/2$, delay becomes **infinite**.

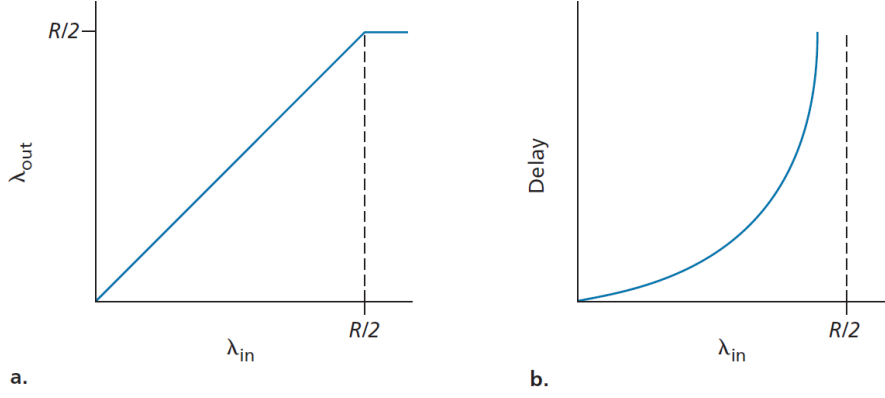


Figure 2: Scenario 1: Throughput and Delay graphs

Cost 1: Queuing Delay

Large queuing delays are experienced as the packet-arrival rate nears the link capacity.

2.2 Scenario 2: Two Senders, Finite Buffers

- **Setup:** Same as Scenario 1, but with **finite router buffers**.
- **Offered Load:** We define λ'_{in} as the rate of original data *plus* retransmitted data.

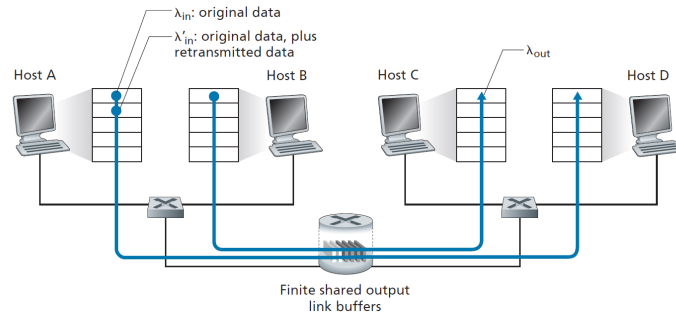


Figure 3: Scenario 2: Finite Buffers and Retransmissions

Performance Analysis

1. **Ideal Case:** Sender only sends when a buffer is free. No loss, $\lambda_{in} = \lambda_{out}$.
2. **Known Loss:** Sender retransmits only when loss is certain. Throughput is lower than $R/2$ because some bandwidth is used for retransmissions.
3. **Premature Timeout:** Sender retransmits packets that are still in queue. Receiver discards duplicates.

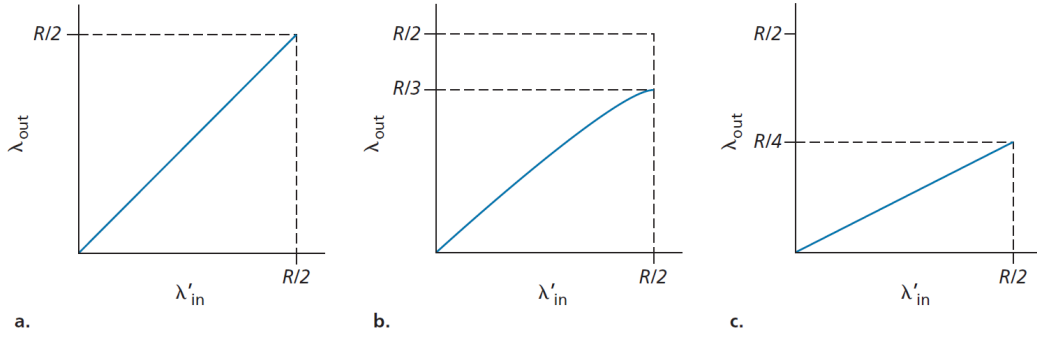


Figure 4: Scenario 2: Comparison of Performance Cases

Cost 2: Retransmission Waste

The sender must perform retransmissions to compensate for dropped packets, and unneeded retransmissions waste link bandwidth.

2.3 Scenario 3: Four Senders, Multihop Paths

- **Setup:** Four hosts send packets over overlapping two-hop paths.
- **Dynamics:** Heavy traffic from one connection (e.g., B–D) can crowd out another (e.g., A–C) at a shared router (R2).

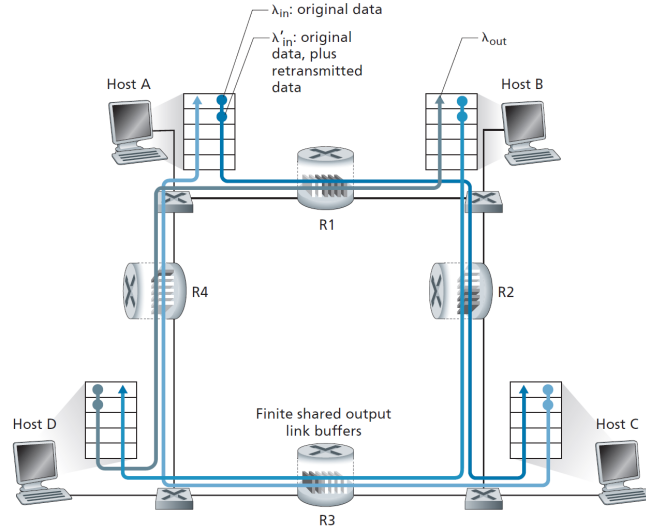


Figure 5: Scenario 3: Multihop Congestion

The "Wasted Work" Effect

When a packet is dropped at the *second* hop, the transmission capacity used at the *first* hop to forward that packet is completely **wasted**. As traffic increases, end-to-end throughput can actually drop to zero.

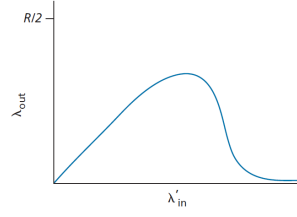


Figure 6: Scenario 3: Throughput Tradeoff

Cost 3: Upstream Resource Waste

When a packet is dropped along its path, the capacity used at each of the upstream links to forward it is wasted.

3. Summary of Congestion Costs

The True Price of Congestion

- **Infinite Delay:** Queuing delays grow exponentially as capacity is reached.
- **Retransmission Burden:** Required to maintain reliability but reduces effective throughput.
- **Inefficiency:** Routers waste work forwarding packets that will eventually be dropped or duplicated.
- **Pathological Throughput:** In multihop networks, high congestion can lead to near-zero global throughput.